

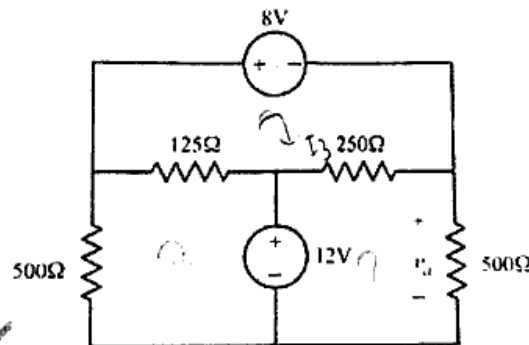
FACULTY OF ENGINEERING

**B.E. (EEE/EIE/CSE/AI/CME/DS) II - Semester (AICTE) (Main & Backlog) (New) Examination,
August / September 2024**

Subject: Basic Electrical Engineering**Time: 3 Hours****Max. Marks: 70**

- Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.
(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.
(iii) Missing data, if any, may be suitably assumed.

1. a) Distinguish between active and passive elements and give examples.
b) What do you mean by impedance?
c) What do you mean by transformation ratio? What is the significance of it?
d) Give reason, why 1- Φ induction motors are not self-starting.
e) A geyser of 2.5kw is used for 8 hours daily. Calculate the monthly consumption (30 days) of electrical energy units. Also calculate the cost of electricity units consumed in a month if rate per unit is Rs.3.50.
f) State Kirchoff's laws.
g) What is phase difference? Justify in a pure inductive circuit voltage leads the current.
2. a) Using mesh analysis find v_a .



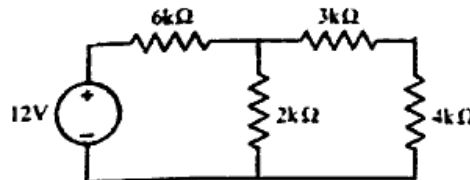
- b) State and explain Norton's theorem with a suitable example.
3. a) Show that in a 3- Φ delta connected system $I_L = \sqrt{3}I_{ph}$.
b) Three circuit elements R, L, C are connected in series. Resistance and reactance of each element is given as $R=10\Omega$, $X_L=12\Omega$ and $X_C=18\Omega$.
(i) Find the drop across each element if the current through the series circuit is 10A.
(ii) If the circuit is operated at 50Hz find the inductance and capacitance value.
(iii) Power factor of the circuit.

4. a) Explain how rotating magnetic field is produced in 3- ϕ induction motor.
 b) A 20kVA, 2500/250V, 50Hz single phase transformer gave the following test result

Test	Voltage (V)	Current(A)	Power(W)
Open Circuit	250	1.4	105
Short Circuit	104	8	320

Determine the efficiency of the transformer when it is operated at

- (i) $\frac{3}{4}$ load at 0.8 pf lag
 (ii) $\frac{1}{2}$ load at 0.6 pf lead
5. a) A 4 pole lap wound DC shunt generator driven by a prime mover with a speed of 1200 rpm delivers 200 A at terminal voltage of 250 V. It has a field and armature resistance of 50Ω and 0.05Ω respectively. Neglecting brush drop determine (i) armature current (ii) EMF generated (iii) flux/pole if the number of armature conductors are 1080.
 b) With a neat sketch explain how capacitor start and capacitor run motor works.
6. a) List the different types of earthing. Explain the earthing methods with a neat sketch.
 b) What is the significance of power factor improvement? Explain any one method to improve the power factor. <https://www.osmaniaonline.com>
7. a) What is a battery? Classify different types of batteries. Compare primary batteries with secondary batteries.
 b) (i) State Thevenin's theorem.
 (ii) Find current in a $4k\Omega$ resistor in the circuit given below using Thevenin's theorem.



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